Diego A. Mesa

damesa@eng.ucsd.edu

EDUCATION

Doctoral Candidate in Bioengineering *December 2015*
University of California, San Diego

Bachelor of Science in Computer Engineering *December 2010*
University of Florida, *Gainesville, FL* (honors)

EXPERIENCE

Graduate Researcher, **Neural Interaction Lab** *August 2011 - Present*

- Developing novel theoretical and numerical techniques that make *efficient* use of *information* and are *scalable* with special applications to the design of global health decision tools.

Undergraduate Research Assistant (REU), University of Florida**,** Gainesville, FL *January 2010 – August 2011*

- Working on a DARPA funded Cyber workstation project with a multidisciplinary team of neuro and computer engineers from different universities to construct a *portable* and *powerful* brain-machine interface workstation capable of running a closed-loop B.M.I experiment with live animal robotic control where the rigorous model computation is performed off-site.

Integrated Product and Process Design, Gainesville, FL *August 2009 – August 2010*

- Working with Raytheon Corp. on a yearlong research and development of a C++ based prototype of a multi-touch video teleconferencing application with military considerations involving an innovative and scalable user interface.

Software Engineering Intern, Lockheed Martin, Orlando, FL *May 2009 – August 2009*

- Worked on the Joint Strike Fighter as a Java developer on the ALIS web interface for fleet management.
- Worked closely with all branches of the military and other customers to ensure the quality of the product.
- Responsible for maintaining an ITAR security awareness and handling sensitive material.

PUBLICATIONS

- S. G. Kim, R. Ma, **D. Mesa**, T. P. Coleman, *"Efficient Bayesian Inference Methods via Convex Optimization and Optimal Transport", IEEE International Symposium on Information Theory (ISIT)*, 2013
- M.J. Harbert , S.S. Rosenberg, **D. Mesa**, M. Sinha, N.P. Karanjia, and T.P. Coleman, *"Demonstration of the Use of Epidermal Electronics in Neurological Monitoring"*, Proceedings of the American Neurological Association Annual Meeting, Oct 2013
- P. Rattanatamrong, A. Matsunaga, P. Raiturkar, **D. Mesa**, M. Zhao, B. Mahmoudi, J. DiGiovanna, J. Principe, R. Figueiredo, J. Sanchez and J. Fortes, *"Model Development, Testing and Experimentation in a CyberWorkstation for Brain-Machine Interface Research", Proceedings of the 32nd Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, 2010

CONFERENCES & TALKS

- **Invited Speaker**, Dept. of Pediatrics – Joint Statistics/Epidemiology Seminar *October 2014*
- Poster, Systems On Nanoscale Information fabriCs (SONIC) Annual Meeting *September 2013*
- Poster, Center for Science of Information (CSOI) Annual Meeting *June 2013*

TEACHING

- Guest Lecturer, Probability and Statistics for Engineers *Fall 2013*
- Teaching Assistant, Thermodynamics *Winter 2013*
- Teaching Assistant, Biostatics *Spring 2012*
- Teaching Assistant, Senior Design *Fall 2012*

SERVICE

- **Graduate Advocate**, Summer Training Academy for Research in the Sciences (STARS) *Sum. 2012, 2013*
    - Served as a graduate advocate for 5-8 underrepresented minority undergraduates participating in the summer research program. Dedicated over 20 hours a week to tutoring, essay writing, graduate and fellowship applications.
- Member, Society for Hispanic Professional Engineers (SHPE) *2006-2011, 2014 – Present*
- Member, Society for Neuroscience (SFN) *2012 – Present*

HONORS

Recipient, NCFDD – Faculty Development Boot-camp Scholarship *Fall 2014*
Recipient, **National Science Foundation – Graduate Research Fellowship** *Fall 2011*
Recipient, UIUC-MERGE Conference Scholarship *Fall 2010*
Recipient, UC-Berkeley EDGE Conference Scholarship *Fall 2010*
Recipient, Summa Cum Laude Graduate from College of Engineering *Fall 2010*
Recipient, College of Engineering – Dean's Scholarship *Spring 2010*
Recipient, Deans List - University of Florida *Fall 2007, Fall 2008, Fall 2009*